

Analysis of Variance (ANOVA)

Analysis of Variance (ANOVA) is a statistical test used to compare the means of **three or more** independent groups to determine if at least one group mean is significantly different from the others.



When to Use ANOVA

ANOVA is essentially an extension of the **t-test** for situations involving more than two groups.

- **Goal:** To determine if the variation **between** the group means is larger than the variation **within** the groups.
- **Outcome Variable:** The variable being measured (e.g., test scores, crop yield) must be a **continuous** variable (interval or ratio data).
- **Factor/Grouping Variable:** The variable that defines the groups (e.g., treatment type, gender, location) must be **categorical**.
- **Alternative to Multiple t-tests:** Instead of running multiple pairwise *t*-tests (which increases the chance of making a Type I error—a false positive), ANOVA provides a single, comprehensive test.



How to Use ANOVA (General Procedure)

ANOVA works by calculating the **F-statistic** ($F_{\text{Calculated}}$), which is the ratio of two different variance estimates:

$$F = \frac{\text{Variance Between Samples (Mean Square Between)}}{\text{Variance Within Samples (Mean Square Within)}}$$

1. **Mean Square Between (MS_B)**: Measures the variability *between* the different group means. It represents the effect of the treatment or factor.
2. **Mean Square Within (MS_W)**: Measures the variability *within* each individual group. It represents random error or chance variation.

If the calculated F -value is **significantly large** (meaning MS_B is much larger than MS_W), we **reject the null hypothesis**, concluding that the groups are not all the same.

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Hypothesis Testing Procedure and Interpretation

Hypotheses

ANOVA always tests the means of the groups being compared:

- **Null Hypothesis (H_0)**: All population means are equal.

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \dots = \mu_k$$

- **Alternative Hypothesis (H_1)**: At least one population mean is different from the others.

$$H_1 : \text{Not all means are equal.}$$

Interpreting ANOVA Results

The final decision is made by comparing $F_{\text{Calculated}}$ to F_{Critical} (or by using the P-value).

Comparison	Result	Interpretation	Next Step
$F_{\text{Calculated}} \leq F_{\text{Critical}}$ (P-value $> \alpha$)	Accept H_0	There is no significant evidence that the group	Stop: Conclude the factor has no effect.

Comparison	Result	Interpretation	Next Step
		means are different.	
$F_{\text{Calculated}} > F_{\text{Critical}}$ (P-value $\leq \alpha$)	Reject H_0	At least one group mean is significantly different.	Proceed to Post-Hoc Tests to find out <i>which</i> specific pairs are different (e.g., Tukey's HSD).



One-Factor vs. Two-Factor ANOVA

The classification of ANOVA depends on the number of **categorical factors** (independent variables) being tested in the model.

1. One-Factor ANOVA (One-Way ANOVA)

- **Definition:** Used when comparing the means of groups created by **one single factor**.
- **Example:** A researcher tests three different teaching methods (Method A, Method B, Method C) on student test scores.
 - *Outcome Variable:* Test Score (Continuous)
 - *Factor:* Teaching Method (Categorical, 3 levels)

2. Two-Factor ANOVA (Two-Way ANOVA)

- **Definition:** Used when comparing means based on **two independent factors** simultaneously.
- **Advantage:** It allows you to test for the separate effect of each factor **and** the **interaction effect** between the two factors.
- **Example:** A researcher tests three different teaching methods (Factor A) and two different class sizes (Factor B: Large, Small) on student test scores.

- **Factor A:** Teaching Method
- **Factor B:** Class Size
- **Interaction:** Does the best teaching method depend on the class size?



Identifying the Appropriate Hypothesis Testing Procedure

The choice of the correct test depends on the nature of the data and the number of samples being compared.

Scenario	Outcome Variable Type	Number of Samples (Groups)	Appropriate Hypothesis Test
Comparing a Sample to a Claim	Continuous	1	Z-test ($n \geq 30$ or σ known) or t-test ($n < 30$ and σ unknown)
Comparing Two Groups	Continuous	2	Z-test (if large samples) or t-test (if small samples)
Comparing Multiple Groups (3 or more)	Continuous	$k \geq 3$	One-Factor ANOVA (or Two-Factor ANOVA, etc.)
Testing Relationship between Categories	Categorical	1 (single contingency table)	Chi-Square Test for Independence
Testing Frequency Distribution	Categorical	1 (single set of frequencies)	Chi-Square Goodness-of-Fit Test

Would you like to move on to solving some examples for One-Factor ANOVA?



ANOVA is used to test the hypothesis that the means of three or more groups are equal. We compare the **variance between groups** to the **variance within groups** using the F-statistic.



Example 1: One-Way ANOVA (Hospital Deaths)

Question: Compare **three Hospitals** (A, B, C) with regards to **number of deaths per month**. Suggest a difference in the no. of the deaths per month among three hospitals. (Assume $\alpha = 0.05$)

Hospital (Group)	Deaths (x)	n_j	$\bar{x}_j$	$\sum x_j$	$\sum x_j^2$
A	3, 4, 3, 5, 0	5	3.0	15	59
B	6, 3, 3, 4, 4	5	4.0	20	74
C	7, 3, 4, 6, 5	5	5.0	25	135
Totals		$N = 15$	$\bar{x}_{\text{Grand}} = 4.0$	$\sum x = 60$	$\sum x^2 = 268$

1. Hypotheses

- $H_0 : \mu_A = \mu_B = \mu_C$ (No difference in mean deaths per month.)
- H_1 : At least one mean is different.

2. Calculate Sums of Squares

A. Total Sum of Squares (SS_T): Measures total variability.

$$SS = \sum x^2 - \frac{(\sum x)^2}{N} = 268 - \frac{(60)^2}{15} = 268 - 240 = \mathbf{28}$$

B. Between-Groups Sum of Squares (SS_B): Measures variability due to the hospitals.

$$SS_B = \sum \frac{(\sum x_j)^2}{n_j} - \frac{(\sum x)^2}{N} = \left(\frac{15^2}{5} + \frac{20^2}{5} + \frac{25^2}{5} \right) - 240$$

$$SS_B = (45 + 80 + 125) - 240 = 250 - 240 = 10$$

C. Within-Groups Sum of Squares (SS_W): Measures variability due to error/chance.

$$SS_W = SS - SS_B = 28 - 10 = 18$$

3. Calculate Mean Squares and F-Ratio

- $k = 3$ groups. $N = 15$ total observations.
- $df_{\text{Between}} (k - 1) = 3 - 1 = 2$
- $df_{\text{Within}} (N - k) = 15 - 3 = 12$
- $MS_B = \frac{SS_B}{df_B} = \frac{10}{2} = 5.0$
- $MS_W = \frac{SS_W}{df_W} = \frac{18}{12} = 1.5$

$$F_{\text{Calculated}} = \frac{MS_B}{MS_W} = \frac{5.0}{1.5} \approx 3.33$$

4. Conclusion

- **Critical Value (F_{Critical}):** For $df = (2, 12)$ and $\alpha = 0.05$, $F_{\text{Critical}} \approx 3.89$ (using common tables).
- $F_{\text{Calculated}} = 3.33$
- Since $3.33 < 3.89$, the calculated F falls within the acceptance region.
- **Result: Accept H_0** (Fail to Reject H_0).
- **Interpretation:** There is **no significant difference** in the mean number of deaths per month among the three hospitals at the 5% level of significance.

Example 2: One-Way ANOVA (Machine Speed)

Question: Determine whether the **three machines (A1, A2, A3)** are **significantly different in their mean speed**. (Given $F_{2,12} = 3.89$)

Machine (Group)	n_j	$\bar{x}_j$	$\sum x_j$	$\sum x_j^2$
A1	5	32.0	160	5174
A2	5	37.0	185	6843
A3	5	29.0	145	4257
Totals	$N = 15$	$\bar{x}_{\text{Grand}} = 32.67$	$\sum x = 490$	$\sum x^2 = 16274$

1. Hypotheses

- $H_0 : \mu_{A1} = \mu_{A2} = \mu_{A3}$ (No difference in mean speed.)
- $H_1 : \text{At least one mean is different.}$

2. Calculate Sums of Squares

A. Total Sum of Squares (SS_T):

$$SS = 16274 - \frac{(490)^2}{15} = 16274 - 16006.67 = \mathbf{267.33}$$

B. Between-Groups Sum of Squares (SS_B):

$$SS_B = \left(\frac{160^2}{5} + \frac{185^2}{5} + \frac{145^2}{5} \right) - 16006.67$$

$$SS_B = (5120 + 6845 + 4205) - 16006.67 = 16170 - 16006.67 = \mathbf{163.33}$$

C. Within-Groups Sum of Squares (SS_W):

$$SS_W = SS - SS_B = 267.33 - 163.33 = \mathbf{104.00}$$

3. Calculate Mean Squares and F-Ratio

- $k = 3$ groups. $N = 15$ total observations.
- $df_{\text{Between}} = 2$

- $df_{\text{Within}} = 12$
- $MS_B = \frac{163.33}{2} \approx 81.67$
- $MS_W = \frac{104.00}{12} \approx 8.67$

$$F_{\text{Calculated}} = \frac{MS_B}{MS_W} = \frac{81.67}{8.67} \approx 9.42$$

4. Conclusion

- **Critical Value (F_{Critical}):** Given as $F_{2,12} = 3.89$.
- $F_{\text{Calculated}} = 9.42$
- Since $9.42 > 3.89$, the calculated F falls into the rejection region.
- **Result: Reject H_0 .**
- **Interpretation:** There is a **highly significant difference** in the mean speed among the three machines. We conclude that at least one machine is significantly faster or slower than the others.

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Assignment Questions Solutions

Assignment 1: One-Way ANOVA (Wheat Varieties)

Question: Test for **significant difference in the production of four varieties of wheat** (A, B, C, D). (The table layout suggests a Two-Way ANOVA, but since the question only asks about the varieties, we proceed with a **One-Way ANOVA** treating Plots as replications).

Variety	n_j	$\sum x_j$	$\bar{x}_j$	$\sum x_j^2$
A	3	630	210	132400
B	3	650	216.67	148900
C	3	695	231.67	196825

Variety	n_j	$\sum x_j$	$\bar{x}_j$	$\sum x_j^2$
D	3	750	250	229500
Totals	$N = 12$	$\sum x = 2725$	$\bar{x}_{\text{Grand}} = 227.08$	$\sum x^2 = 707625$

- F_{Critical} for varieties: $F_{3,8} = 4.07$ (since $k - 1 = 3$ and $N - k = 8$).

1. Calculate Sums of Squares

- **Correction Factor (CF):** $\frac{(\sum x)^2}{N} = \frac{2725^2}{12} \approx 618,802.08$
- $SS_T = 707625 - 618802.08 = \mathbf{88,822.92}$
- $SS_B = \left(\frac{630^2}{3} + \frac{650^2}{3} + \frac{695^2}{3} + \frac{750^2}{3} \right) - 618802.08$

$$SS_B = (132300 + 140833.33 + 160958.33 + 187500) - 618802.08 = 621591.66 - 618802.08 =$$

- $SS_W = SS - SS_B = 88822.92 - 2789.58 = \mathbf{86,033.34}$

2. Calculate Mean Squares and F-Ratio

- $MS_B = \frac{SS_B}{k-1} = \frac{2789.58}{3} \approx \mathbf{929.86}$
- $MS_W = \frac{SS_W}{N-k} = \frac{86033.34}{8} \approx \mathbf{10,754.17}$

$$F_{\text{Calculated}} = \frac{MS_B}{MS_W} = \frac{929.86}{10754.17} \approx \mathbf{0.0865}$$

3. Conclusion

- $F_{\text{Calculated}} = 0.0865$
- F_{Critical} for (3, 8) d.f. = **4.07**.
- Since $0.0865 < 4.07$, **Accept** H_0 .
- **Answer:** There is **no significant difference** in the production of these four varieties of wheat.

Assignment 2: One-Way ANOVA (Wheat Varieties A, B, C)

Question: Test for **significant difference in the production of three varieties A, B and C**. (Given $F_{2,9} = 4.261$)

Variety	Data (x)	n_j	$\sum x_j$	$\sum x_j^2$
A	14, 16, 18	3	48	776
B	14, 13, 15, 22	4	64	1086
C	18, 16, 19, 19, 20	5	92	1702
Totals		$N = 12$	$\sum x = 204$	$\sum x^2 = 3564$

- F_{Critical} for varieties: $F_{2,9} = 4.261$ (since $k - 1 = 2$ and $N - k = 9$).

1. Calculate Sums of Squares

- Correction Factor (CF):** $\frac{(204)^2}{12} = 3468$
- $SS_T = 3564 - 3468 = \mathbf{96}$
- $SS_B = \left(\frac{48^2}{3} + \frac{64^2}{4} + \frac{92^2}{5} \right) - 3468$

$$SS_B = (768 + 1024 + 1692.8) - 3468 = 3484.8 - 3468 = \mathbf{16.8}$$

- $SS_W = 96 - 16.8 = \mathbf{79.2}$

2. Calculate Mean Squares and F-Ratio

- $df_{\text{Between}} = 2$
- $df_{\text{Within}} = 9$
- $MS_B = \frac{16.8}{2} = \mathbf{8.4}$
- $MS_W = \frac{79.2}{9} = \mathbf{8.8}$

$$F_{\text{Calculated}} = \frac{MS_B}{MS_W} = \frac{8.4}{8.8} \approx \mathbf{0.9545}$$

3. Conclusion

- $F_{\text{Calculated}} = 0.9545$

- F_{Critical} for (2, 9) d.f. = **4.261**.
- Since $0.9545 < 4.261$, **Accept H_0** .
- **Answer:** There is **no significant difference** in the production of the three varieties of wheat (A, B, and C).

Would you like to review the concepts of **Control Charts** now, or perhaps explore a different area of statistics?